KITTI Splat Architecture

High-Fidelity 3D Street Reconstruction

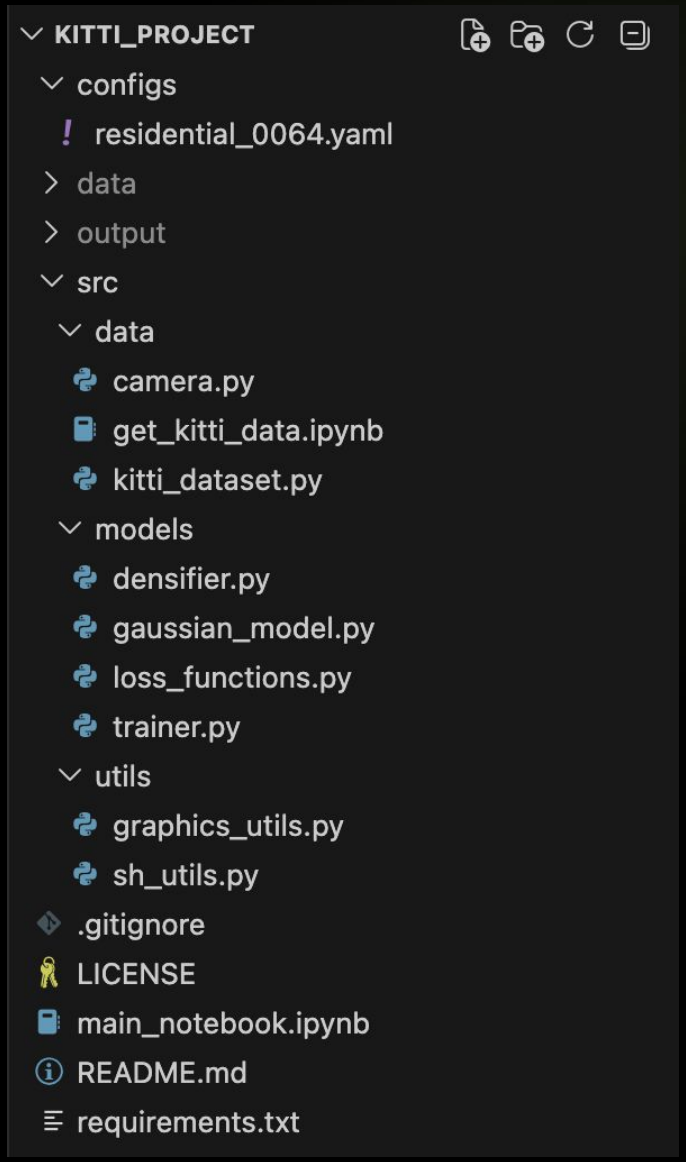
SOFTWARE SYSTEM DESCRIPTION

System Execution Flow

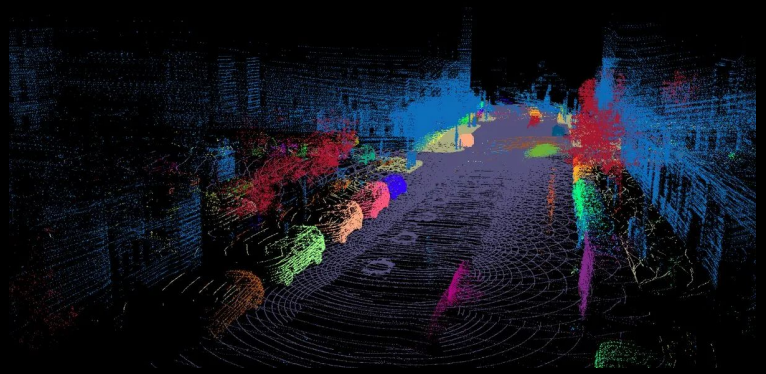
End-to-End Orchestration

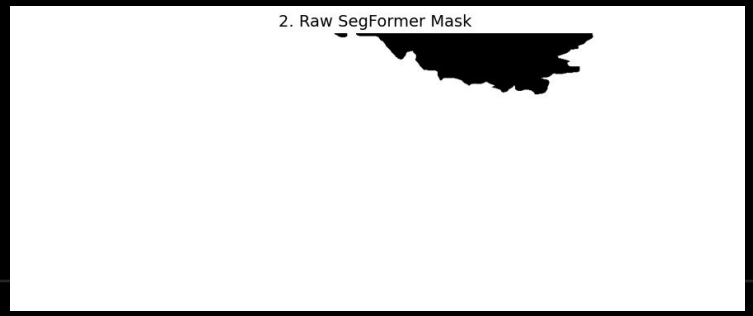
The architecture is a modular pipeline that transforms **raw sensor data** into an **explicit 3D world model**

- **Input:** Synced LiDAR, Camera, & OXTS GPS
- **Process:** Gaussian Optimization & Guardrails
- **Output:** Photorealistic Real-time Navigable PLY



Module: KITTI Dataset Parser





src/data/kitti_dataset.py

Handles multi-modal ingestion with temporal synchronization.

- Loads rectified Image_02 (RGB)
- Parses Velodyne .bin LiDAR scans
- Integrates SegFormer Semantic Masks

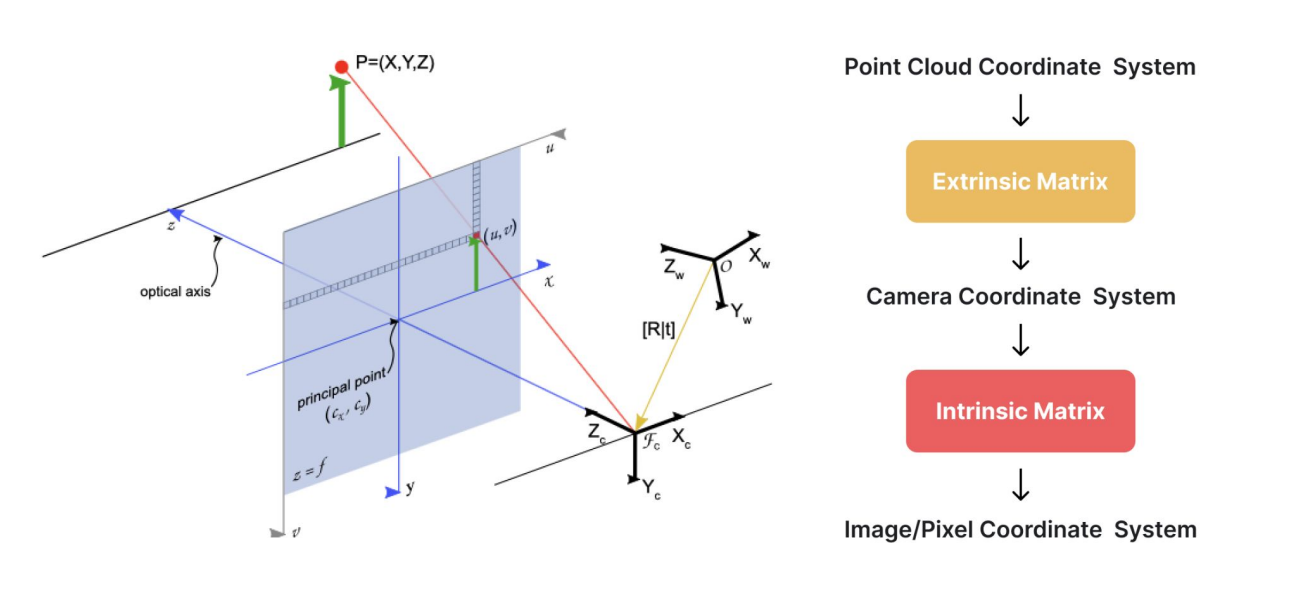
Sensor Fusion Logic

src/data/camera.py

This module implements the geometric bridge between 3D world space and 2D pixel space.

Utilizes 6-DOF extrinsic matrices and projection parameters to align Velodyne points with RGB pixels.

- Velo-to-Cam calibration mapping
- World-to-Velo rigid transformation
- Perspective intrinsic matrix handling



Model: The Gaussian Primitives

Geometry

Mean position (XYZ) and Covariance defined by Scaling (S) and Rotation (Q).

Appearance

Spherical Harmonics (SH) coefficients for view-dependent color rendering.

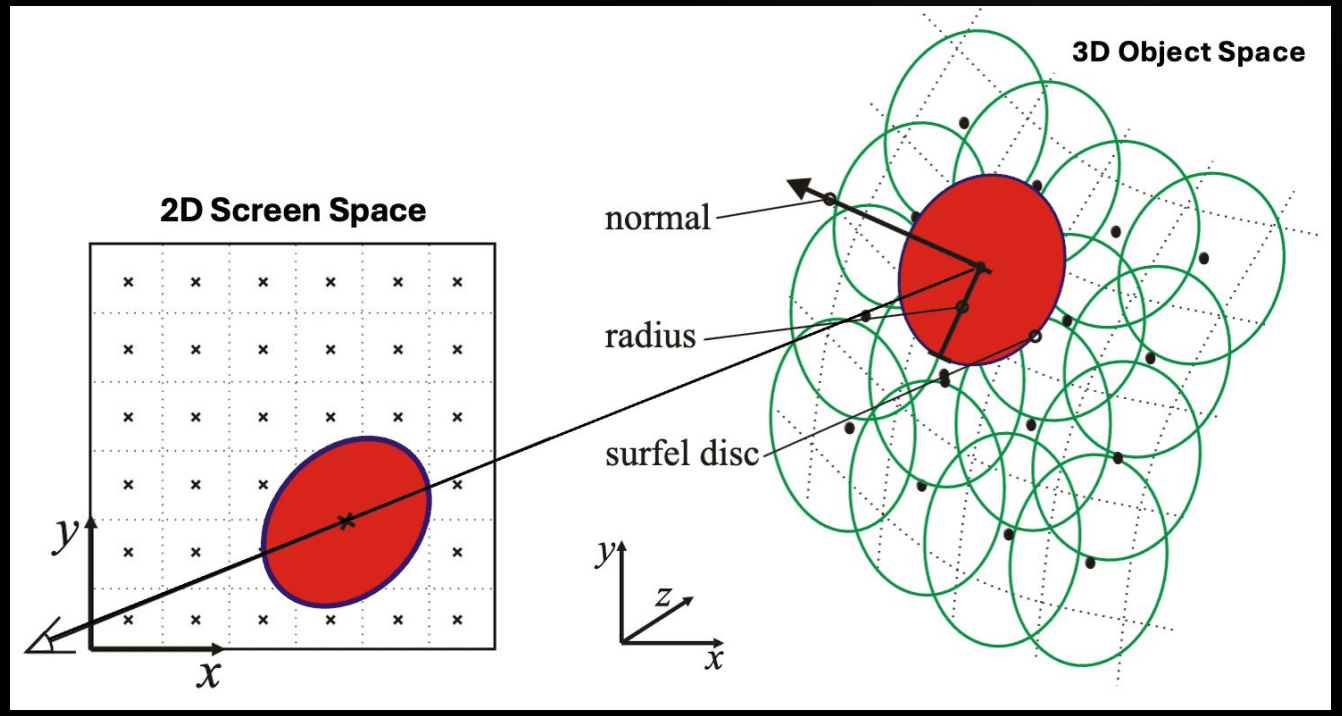
Opacity

Learned alpha values (0-1) defining the transparency of each ellipsoid.

Explicit State

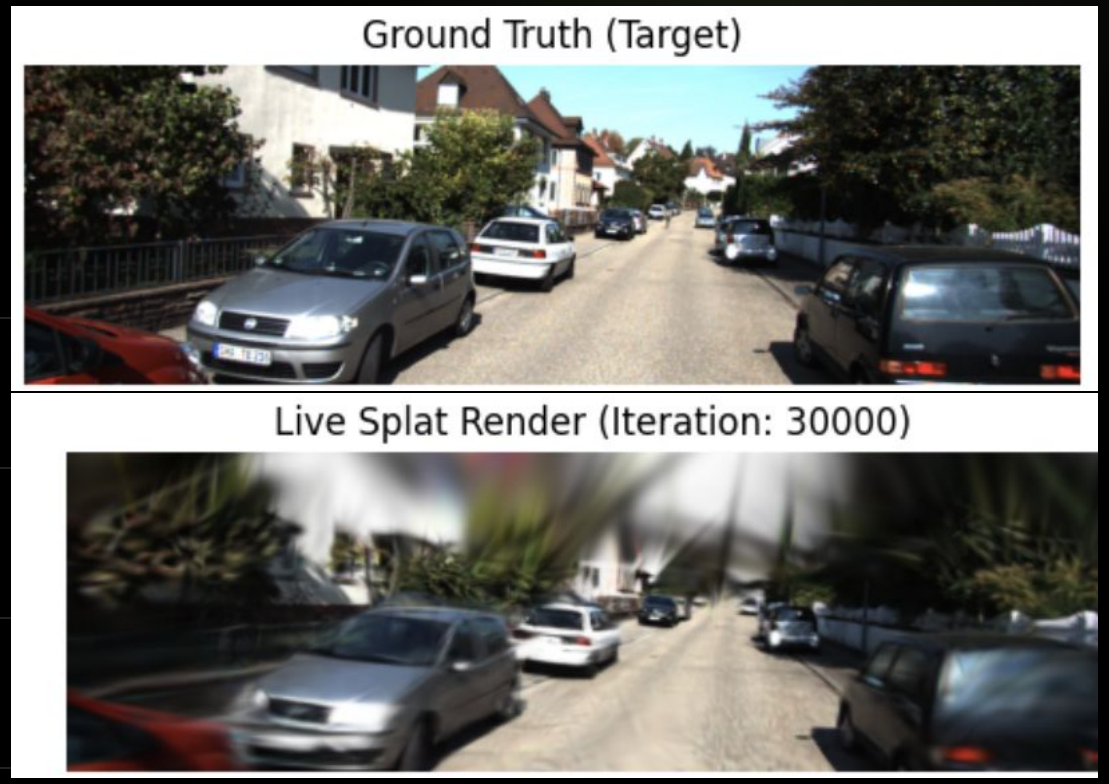
Parameters stored as tensors for efficient CUDA gradient backpropagation.

src/models/gaussian_model.py



Orchestration: Optimization Loop

File: src/models/trainer.py	Functionality Description
Iteration Loop	30,000 steps of stochastic gradient descent optimization.
Densification	Adding Gaussians to under-reconstructed areas (gradient threshold).
Pruning	Eliminating low-contribution Gaussians to reduce memory.
Baking	Final high-res refinement after densification shut-off (at 25k).



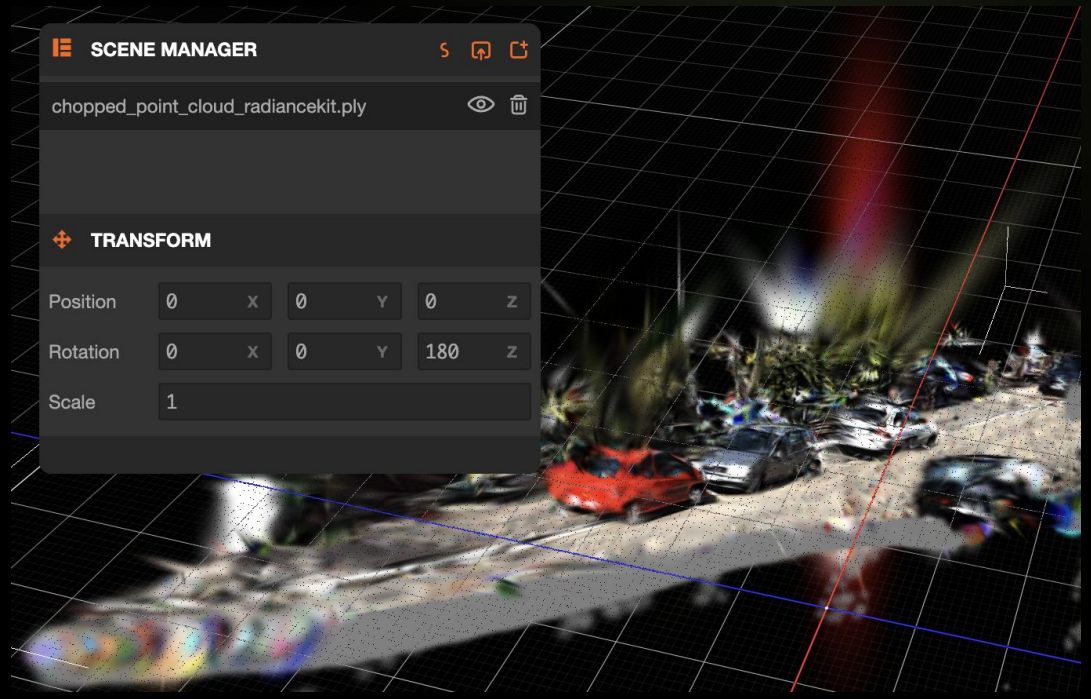
System Guardrails & Constraints

src/models/densifier.py

Proprietary logic developed to solve KITTI's unique artifact challenges:

> **Scale Clamp (1.5m):** Prevents Gaussians from growing into giant needles, eliminating the "sky spears."
>
> **Hard Height Limit:** Instantly deletes any point that grows above 8 meters, ensuring stable street geometry.

Navigable Output

The final software output is an explicit .PLY file packaged for the SuperSplat web visualizer.

- Point Cloud Export
- SuperSplat Browser Navigable Interface